

The bionomics of *Anopheles arabiensis* and *Anopheles funestus* inside local houses and their implications for vector control strategies in areas with high coverage of insecticide-treated nets in South-eastern Tanzania.

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Residual malaria transmissions in Africa may be associated with improved coverage of insecticide-treated nets, house features, and livestock husbandry. These human-land use activities may drive the ecology and behaviour of malaria vectors which sustain residual malaria transmission. This study was conducted to assess changes in the ecology and behaviour of *Anopheles funestus* and *Anopheles arabiensis* in villages with high coverage of insecticide-treated nets to guide the selection of complementary vector control strategies against residual malaria transmission. Mosquitoes were collected using a CDC-light trap, miniaturized double net trap, and Prokopack aspirator from 222 households in three villages (Ebuyu, Chirombora, and Mzelezi) within Kilombero Valley. *Anopheles* mosquitoes were morphologically identified to their physiological status and species-complex levels. A sub-sample of *Anopheles* mosquitoes was exposed to laboratory analyses of sibling species, host preference, and sporozoite rates. Additionally, the local houses were geo-referenced using Global Positioning Systems (GPS) device, and house features were recorded and associated with vector abundance. The population of *An. funestus* s.s was abundant with high *Plasmodium* sporozoite rates inside houses compared to *An. arabiensis*. However, these vector species equally blood-fed on humans inside houses, but they also flexibly mixed human and animal blood meal. Fewer *An. funestus* were caught in houses with metal- than grass roofs and houses with and without animals. Contrastingly, fewer *An. arabiensis* were caught from houses with screened eaves compared to houses with open eaves. This study confirms that *An. funestus* dominates residual malaria transmission over *An. arabiensis*. These vector species exhibit anthropophily and opportunistic blood-feeding behaviour in areas with high coverage of insecticide-treated nets, but they numerically respond differently to local house improvements. These results imply that integrating mosquito-proof houses, improved insecticide-treated nets, and livestock-based interventions could effectively reduce and eventually eliminate residual malaria transmission.

Mapping *Plasmodium* transitions and interactions in the *Anopheles* female.

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The human malaria parasite, *Plasmodium falciparum*, relies exclusively on *Anopheles* mosquitoes for transmission. Once ingested during blood feeding, most parasites die in the mosquito midgut lumen or during epithelium traversal¹. How surviving ookinetes interact with midgut cells and form oocysts is poorly known, yet these steps are essential to initiate a remarkable growth process culminating in the production of thousands of infectious sporozoites². Here, using single-cell RNA sequencing of both parasites and mosquito cells across different developmental stages and metabolic conditions, we unveil key transitions and mosquito-parasite interactions occurring in the midgut. Functional analyses uncover processes regulating oocyst growth and identify the transcription factor PfSIP2 as essential for sporozoite infection of human hepatocytes. Combining shared mosquito-parasite barcode analysis with confocal microscopy, we reveal that parasites preferentially interact with midgut progenitor cells during epithelial crossing, potentially using their basal location as an exit landmark. Additionally, we show tight connections between extracellular late oocysts and surrounding muscle cells that may ensure parasites adherence to the midgut. We confirm our major findings in several mosquito-parasite combinations, including field-derived parasites. Our study provides fundamental insight into the molecular events

characterizing previously inaccessible biological transitions and mosquito-parasite interactions, and identifies candidates for transmission-blocking strategies.

Molecular characterization of Anopheline (Diptera: Culicidae) mosquitoes from eight geographical locations of Sri Lanka.

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Genus *Anopheles* is a major mosquito group of interest in Sri Lanka as it includes vectors of malaria and its members exist as species complexes. Taxonomy of the group is mainly based on morphological features, which are not conclusive and can be easily erased while handling the specimens. A combined effort, using morphology and DNA barcoding (using the markers cytochrome c oxidase subunit I (COI) gene and internal transcribed spacer 2 (ITS2) region, was made during the present study to recognize anophelines collected from eight districts of Sri Lanka for the first time. Cytochrome c oxidase subunit I and ITS2 regions of morphologically identified anopheline mosquitoes from Sri Lanka were sequenced. These sequences together with GenBank sequences were used in phylogenetic tree construction and molecular characterization of mosquitoes. According to morphological identification, the field-collected adult mosquitoes belonged to 15 species, i.e., *Anopheles aconitus*, *Anopheles annularis*, *Anopheles barbirostris*, *Anopheles culicifacies*, *Anopheles jamesii*, *Anopheles karwari*, *Anopheles maculatus*, *Anopheles nigerrimus*, *Anopheles pallidus*, *Anopheles peditaeniatus*, *Anopheles pseudojamesi*, *Anopheles subpictus*, *Anopheles tessellatus*, *Anopheles vagus*, and *Anopheles varuna*. However, analysis of 123 COI sequences (445 bp) (16 clades supported by strong bootstrap value in the neighbour joining tree and inter-specific distances of >3%) showed that there are 16 distinct species. Identity of the morphologically identified species, except *An. subpictus*, was comparable with the DNA barcoding results. COI sequence analysis showed that morphologically identified *An. subpictus* is composed of two genetic entities: *An. subpictus* species A and species B (inter-specific K2P distance 0.128). All the four haplotypes of *An. culicifacies* discovered during the present study belonged to a single species. ITS2 sequences (542 bp) were obtained for all the species except for *An. barbirostris*, *An. subpictus* species B, *An. tessellatus*, and *An. varuna*. Each of these sequences was represented by a single species-specific haplotype. The present study reflects the importance and feasibility of COI and ITS2 genetic markers in identifying anophelines and their sibling species, and the significance of integrated systematic approach in mosquito taxonomy. Wide distribution of malaria vectors in the country perhaps indicates the potential for re-emergence of malaria in the country.

Non-coding RNA gene families in the genomes of anopheline mosquitoes.

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Only a small fraction of the mosquito species of the genus *Anopheles* are able to transmit malaria, one of the biggest killer diseases of poverty, which is mostly prevalent in the tropics. This diversity has genetic, yet unknown, causes. In a further attempt to contribute to the elucidation of these variances, the international "Anopheles Genomes Cluster Consortium" project (a.k.a. "16 Anopheles genomes project") was established, aiming at a comprehensive genomic analysis of several anopheline species, most of which are malaria vectors. In the frame of the international consortium carrying out this project our team studied the genes encoding families of non-coding RNAs (ncRNAs), concentrating on four classes:

microRNA (miRNA), ribosomal RNA (rRNA), small nuclear RNA (snRNA), and in particular small nucleolar RNA (snoRNA) and, finally, transfer RNA (tRNA). Our analysis was carried out using, exclusively, computational approaches, and evaluating both the primary NGS reads as well as the respective genome assemblies produced by the consortium and stored in VectorBase; moreover, the results of RNAseq surveys in cases in which these were available and meaningful were also accessed in order to obtain supplementary data, as were "pre-genomic era" sequence data stored in nucleic acid databases. The investigation included the identification and analysis, in most species studied, of ncRNA genes belonging to several families, as well as the analysis of the evolutionary relations of some of those genes in cross-comparisons to other members of the genus *Anopheles*. Our study led to the identification of members of these gene families in the majority of twenty different anopheline taxa. A set of tools for the study of the evolution and molecular biology of important disease vectors has, thus, been obtained.

Ecological meta-analysis of density-dependent processes in the transmission of lymphatic filariasis: survival of infected vectors.

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The survival rate of infected vectors represents one of the fundamental components that influence the transmission dynamics of mosquito-borne diseases. Despite the occurrence of a number of studies investigating mosquito survival after infection with filarial worms, there remains conflicting evidence from both laboratory and field experiments as to the existence and mechanism for parasite-induced mortality among filarial mosquitoes. Here, we used a mixed effects meta-analytical framework to combine the data from all available vector-human host blood feeding experiments to evaluate the evidence for the impact of parasite load on the mortality rates of the three major lymphatic filariasis transmitting mosquito genera, *Culex*, *Aedes*, and *Anopheles* mosquitoes, over the extrinsic incubation period of parasitic infection. The results show that, despite the application of this approach, or in the case of *Anopheles* using a convention fixed effects logistic regression analysis supplemented with additional survival analysis of longitudinal data, no strong association between mortality rate and microfilariae (mf) uptake for either of the three mosquito genera is apparent in the combined data. Instead, a key finding is that study effects played a more crucial role in determining the levels of mortality observed in these experimental studies. This was most revealing in the case of *Culex*, given that the largest single study in terms of both the number of data points and range of mf intensities, in contrast to smaller studies, showed a significant positive association between mf intensity and mortality, indicating that in this genus at least, the detrimental effect of infection may be manifested only at the highest mf intakes. Although no density dependence in vector mortality was also observed for *Aedes*, possibly because of the use of restricted human mf intensity range in previous studies, an intriguing finding was that a significantly higher overall mortality was observed for this genus over mfintake ranges that produced much less corresponding mortality in *Culex* and *Anopheles*. The results also indicate that currently very little can be said about the survival rate of *Anopheles* mosquitoes infected with filarial worms because of the striking paucity of data for this genus. Further studies, using standardized methods and covering an appropriate range of mf uptake intensities and using study frameworks that allow the design and comparison of data from both experimental and field experiments, are clearly indicated if we are to reliably quantify the likely effect of filarial infection on vector survival.

Species Composition, Parous Rate, and Infection Rate of *Anopheles* Mosquitoes

(Diptera: Culicidae) in Bahir Dar City Administration, Northwest Ethiopia.

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Anopheles mosquitoes are the main vectors of malaria. There is little information on the current entomological aspects of Anopheles mosquitoes in Amhara region of northwestern Ethiopia. Therefore, the aim of this study was to assess the prevailing species composition, parous rate, and infection rate of Anopheles mosquitoes in the Bahir Dar city administration. A community-based cross-sectional study was conducted from January through July 2020. For this, six Centers for Disease Control and Prevention light traps (three traps indoor and three traps outdoor) were used to collect adult female Anopheles mosquitoes. The species were morphologically identified, and the parous and infection rates were determined via dissection of ovaries and salivary gland, respectively. A total of 378 adult female Anopheles mosquitoes comprised of three species (Anopheles d'thali, Anopheles rhodesiensis, and Anopheles gambiae complex) were collected and identified at the study sites. Anopheles rhodesiensis was the predominant species accounting for 90% of all collections at the Zenzelima site, followed by An. gambiae complex (6.5%). In contrast, An. gambiae complex was the predominant species at the Tis Abay site, comprising 94% of captures. The overall parous and infection rates were 35 (62.5%) and 1 (2.9%), respectively.

Olfaction in Anopheles mosquitoes.

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As vectors of disease, mosquitoes are a global threat to human health. The Anopheles mosquito is the deadliest mosquito species as the insect vector of the malaria-causing parasite, which kills hundreds of thousands every year. These mosquitoes are reliant on their sense of smell (olfaction) to guide most of their behaviors, and a better understanding of Anopheles olfaction identifies opportunities for reducing the spread of malaria. This review takes a detailed look at Anopheles olfaction. We explore a range of topics from chemosensory receptors, olfactory neurons, and sensory appendages to behaviors guided by olfaction (including host-seeking, foraging, oviposition, and mating), to vector management strategies that target mosquito olfaction. We identify many research areas that remain to be addressed.

[Supplement to the list of Anopheles (Diptera, Culicidae) mosquitoes of Tadjikistan and the predominant types of vectors in the current foci of malaria in the republic].

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The species composition of and the indices of predominance of Anopheles mosquitoes were determined in the current foci of malaria in the Republic of Tadjikistan. The predominant types of malaria mosquitoes were An. superpictus, An. pulcherrimus, and An. hyrcanus were found to be in the Khatlon Region of Tadjikistan and in the contiguous areas of Uzbekistan. An. superpictus prevailed in the study foci of malaria and, as An. pulcherrimus, were the most dangerous vectors of malaria in the south of the republic. The main vectors of malaria were shown to be An. maculipennis sensu stricto and An. superpictus in the environs of the town of Khudjand, Sogdi Region (Tadjikistan). An. maculipennis was first recorded for Tadjikistan and should be included into the list of malaria mosquitoes of the republic. It is suggested that An. pulcherrimus plays an important role in the spread of malaria in the north of

Tadjikistan.

Midgut microbiota of the malaria mosquito vector *Anopheles gambiae* and interactions with *Plasmodium falciparum* infection.

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The susceptibility of *Anopheles* mosquitoes to *Plasmodium* infections relies on complex interactions between the insect vector and the malaria parasite. A number of studies have shown that the mosquito innate immune responses play an important role in controlling the malaria infection and that the strength of parasite clearance is under genetic control, but little is known about the influence of environmental factors on the transmission success. We present here evidence that the composition of the vector gut microbiota is one of the major components that determine the outcome of mosquito infections. *A. gambiae* mosquitoes collected in natural breeding sites from Cameroon were experimentally challenged with a wild *P. falciparum* isolate, and their gut bacterial content was submitted for pyrosequencing analysis. The meta-taxogenomic approach revealed a broader richness of the midgut bacterial flora than previously described. Unexpectedly, the majority of bacterial species were found in only a small proportion of mosquitoes, and only 20 genera were shared by 80% of individuals. We show that observed differences in gut bacterial flora of adult mosquitoes is a result of breeding in distinct sites, suggesting that the native aquatic source where larvae were grown determines the composition of the midgut microbiota. Importantly, the abundance of Enterobacteriaceae in the mosquito midgut correlates significantly with the *Plasmodium* infection status. This striking relationship highlights the role of natural gut environment in parasite transmission. Deciphering microbe-pathogen interactions offers new perspectives to control disease transmission.

Insecticide resistance in *Anopheles* mosquitoes : additional obstacles in the battle against malaria.

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Mosquitoes of the genus *Anopheles* are malaria vectors in tropical areas and were of course designated as primary targets by programs for malaria control. Repellent sprays, indoor use of insecticides, and massive delivery of mosquito nets are standard examples of the means widely used to combat malaria. This synthetic review supplies an overview of all of the modes of resistance developed by *Anopheles* mosquitoes against these human actions. The misuse of each available tool has gradually led to a decrease in its global effectiveness. Newly-emerging forms of resistance, due to modification or overexpression of molecular targets, as well as behavioral adaptations by mosquitoes, are some examples of the consequences. To enable a categorical reduction in malaria incidence, a thorough adjustment of the use of the various means of control should be envisioned.